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Spring Boot and Microservices
Duration: 40 Hours

TOC version: 1.0

Day No	Session No	Learning Time (in hours)	Module Name	Topic	Sub-Topics	Remarks
Day 01	Session 1	1	Spring Boot and Data JPA	Project Briefing	**Quick Overview:** **Agenda & Objective**: Introducing REST API development with Spring Boot Module and transforming the same to Microservices Architecture **Project Overview & Activities:** a. Present Business scenario - Building REST API (e.g., a simple real-time problem) b. Functional Requirements break down c. Discussion on Technical Pre-requisites d. Walkthrough of REST API Principles	Faculty to give an overview and explain the important aspects of these topics from implementation perspective through hands-on
		0.5		Spring Boot (SB) Overview	**Quick Overview:** Spring Boot benefits over MVC and Core, Setting and Running Sample Boot Project, Starter Dependencies	
		1.5		SB Project Creation and Deep Dive	**Project Task**: Create a Sample Spring Boot and run it with basic needed starter dependencies **Activities:** a. Overview of Spring Boot and its advantages b. Setting up a sample Spring Boot project c. Understanding @SpringBootApplication and @EnableAutoConfiguration d. Introduction to application.properties or application.yml file for custom configuration. e.Custom configuration with application.yml file, e.g., Demo Port Change using Properties and ML File f. Walkthrough of Starters: Pre-defined spring-boot-starter-parent & default Starter Dependencies g. Examples of commonly used Starter Dependencies, such as spring-boot-starter-web for web applications, spring-boot-starter-data-jpa for Spring Data JPA with Hibernate, etc.	
		1		Creating SB Apps for Business scenario	**Project Task**: Create and do the needed Initialization of respective SB Applications needed for implementing the Business scenario functionalities **Activities:** As per the chosen Project/Business Scenario for e.g., Order Management System / BookStore Management System / Employee Management System a. Instantiate new Spring Boot project Apps, Create SB Apps that must further be worked upon b. Later transform them into the Rest API Endpoint c. Configure project dependencies for a RESTful API.	
	Session 2	0.5		Rest Endpoint Overview	**Quick Overview -** Create RESTful API Endpoints using Annotations: @RestController-@RequestBody: produces-consumes, @ResponseBody, ResponseEntity and Test the Endpoints with with a REST Client	
		1		Request Handlers	**Project Task**: Create the needed Request Handlers **Activities:** a. Create the needed Controllers with Request Handlers/methods as per the functionalities in the chosen Project b. Ensure that all the Request Handlers are present with the respective action of - Create - Read - Update - Delete	
		1.5		REST Controllers & Endpoint	**Project Task**: Define and implement the REST API Endpoints by **Activities:** a. Transforming the Controllers to RestControllers with the appropriate @RestController and @RequestMapping annotations for request handling in the chosen Project/Business Scenario b. Implement HTTP operations (GET, POST, PUT, DELETE) using appropriate mappings (@GetMapping, @PostMapping, etc.).	
		1		Annotations DeepDive	**Project Task:** Make use of the other important annotation **Activities:** Ensure the Implementation and usage of @RequestBody: To handle incoming data from the request that will be leveraged/used for adding and updating entity details in the storage @ResponseBody: Transform the RequestHandlers to return an entity in the HTTP response body. based on the HTTP action performed and the Business Scenario handled ResponseEntity: Create a method that returns a ResponseEntity with the Entity details, including status codes and headers.	
		0.5		Models	**Project Task:** Implementing the Models **Activities:** a. Identify the Models that are needed for each of the Functionality b. Explain the importance of the Data Models c. Instantiate/Create the needed Models in the respective project directories	
		1		API Endpoints Testing	**Project Task**: Test the API endpoints developed for the respective Business Scenarios using a REST Client (E.g., Postman, ThunderClient, etc.) just to check if the requests are reaching/hitting the respective Endpoints before proceeding with the integration of Data JPA **Activities:** a. Send GET requests to retrieve and view all entity records b. Send POST requests to add a new entity record to the storage c. Send PUT requests to update the existing details of an entity record d. Send DELETE requests to delete any specific or all entities. e. Verify responses, including status codes, headers, and body.	
		0.5		Spring Data JPA Overview	**Quick Overview -** DataJPA: ORM Overview, Data JPA Starter dependency, Entities, JPA related Annotations: @Entity-@Table-@Id-@Column-@GeneratedValue	

Day 02	Session 1	1		Spring Data JPA - Architecture	**Project Task**: Implement a data layer for the specific Project **Activities:** a. Illustrate the architecture of Object-Relational Mapping (ORM) Framework and its role in abstracting database interactions into an object-oriented model by introducing Java Persistence API (JPA) as a specification for ORM in Java. b. Understand Data JPA and ORM concepts. c. Configure the Spring Boot projects with the Data JPA Starter dependency. d. Create JPA entities for the respective data models e. Define repository interfaces for data access	Faculty to give an overview and explain the important aspects of these topics from implementation perspective through hands-on
		1		Spring Data JPA - Repositories	**Project Task:** Implement Repositories **Activities:** a. Identify the Repositories that have to be created as a part of the Project for the modules/functionalities that need interaction with the Database for data persistence b. Leverage the stereotype annotations @Repository for representation c. Finally, create repository interfaces extending JpaRepository for data access and taking into consideration above mentioned points d. Implement queries as mentioned in the "Querying data" section for accessing the data from the DB with various methods from the JPARepository	
	Session 2	1		Spring Data JPA - Entity Annotations	**Project Task**: Transform the Data Models into Entities **Activities:** a. Use the Annotations that are needed to transform the Data Models into Entities b. Implement JPA entities with annotations: @Entity, @Table, @Id, @Column, @GeneratedValue. c. Explain the importance of annotations in designing the Database Schema	
		1		Spring Data JPA - DB Configuration	**Project Task:** Configure database settings for the Project considered **Activities:** a. Set database URL, username, password, and JPA-specific properties in application.properties or application.yml. b. Integrate repositories into service classes for data manipulation.	
		1		Spring Data JPA - Querying data	**Project Task:** Implement data querying for the Project Functionalities **Activities:** a. Leverage Query by method name: Implement methods like findByTitle, findByAuthorName. b. Leverage the option of Custom Queries with @Query c. Define custom JPQL or SQL queries for complex data retrieval. (optional) d. Test the queries if they are working as expected with the REST API Calls using any Rest Client tool	
		1		End-to-End Testing	**Project Task:** Testing the Endpoints **Activities:** Ensure that all the endpoints of all the modules/functionalities developed as a part of the Project are working well as expected	
Day 03	Session 1	1	Microservices	Project Briefing	**Quick Overview:** **Agenda & Objective**: Introduce Microservices Architecture and implement it leveraging Spring Cloud features, transforming the Spring Boot Applications developed in Day-2 to Microservices **Project Overview & Activities:** a. Present Business scenario - Building Microservices (e.g., a simple real-time problem) b. Functional Requirements break down c. Discussion on Technical Pre-requisites d. Walkthrough of the the importance of Microservices Architecture	Faculty to give an overview and explain the important aspects of these topics from implementation perspective through hands-on
		1		Microservices - Architecture Overview and Benefits	**Project Task**: Transforming SB Apps to Microservices **Activities:** a. Objective: Implement the transition of the Rest API developed with Spring Boot Applications for the considered project to a microservices architecture. b. Understand the differences between monolithic and microservices architectures. c. Discuss the advantages and challenges of microservices.	
		1		Service Registration and Discovery	**Project Task**: Implement service registry and discovery for the bookstore microservices. **Activities:** a. Set up Eureka server and clients. b. Register services with Eureka and demonstrate service discovery.	
		1		API Gateway	**Project Task**: Implement an API Gateway for the bookstore microservices. **Activities:** a. Set up Spring Cloud Gateway. b. Configure route predicates and define custom filters. c. Direct API traffic to appropriate microservices. d. Test the Microservices (Endpoints) accessibility with the API Gateway	
	Session 2	4		Inter-Microservice Communication	**Project Task**: Implement communication between the bookstore microservices. **Activities:** a. Configure RestTemplate for API calls. b. Demonstrate API calls using RestTemplate. c. Introduce OpenFeign for declarative REST client. d. Test the Microservice communication using the Open Feign Client Approach Developed	
Day 04	Session 01	4	Assignment	Assignment + Track-End Quiz Assessment		
	Session 02	4	Assignment	Assignment + Track-End Quiz Assessment		
Day 05	Session 01	4	Assignment	Assignment + Track-End Quiz Assessment		
	Session 02	4	Assignment	Assignment + Track-End Quiz Assessment		